

Retention Strategy Business Proposal

Comprehensive Data Approaches to Predicting Workforce Attrition

Table of Content

1. Executive Summary
2. The Business Problem
3. Impact of Attrition
4. Approach and Goals
5. Methodology
6. Model Performance & Key Findings
7. Recommendations
8. References

Executive Summary

This proposal outlines a data-driven strategy for reducing employee attrition. By analyzing all available organizational data, the **Gradient Boosting Classifier** achieved 87.53% accuracy and a **ROC-AUC** score of **85%**, reflecting its predictive ability for employee attrition. This proposal outlines intervention program to proactively address these issues, reduce turnover and save costs.

The Business Problem: Attrition



Increased Costs

Hiring new employees often involves in advertising, recruiting, interviewing and training which can be expensive



Loss of Knowledge

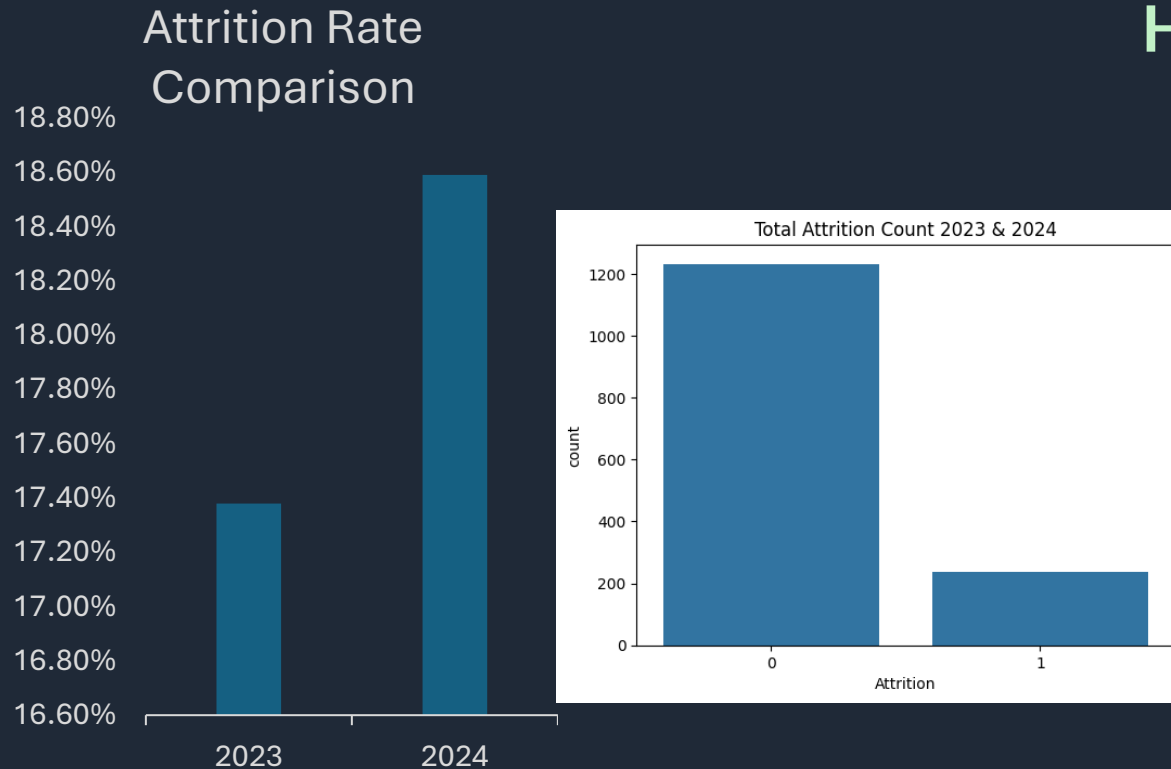
Experienced employees hold institutional knowledge and technical expertise, impacting ongoing projects and efficiency



Reduced Productivity

Turnover increase the workload of remaining employees, while new recruit employees take time to be trained

Impact of Attrition



How big is the costs?

The Attrition Rate is calculated as below:

$$\text{Attrition Rate} = \frac{\text{Number of Employees Who Left}}{\text{Average Number of Employees}} \times 100\%$$

Source 1: Siocon (2024)

- A healthy organizations have an attrition rate of **10% or less**
Source 2: Matthew (2025)
- Average costs of replacing an Employee in UK is around **£25,000** per worker, even up to **£100,000** in senior roles
Source: 3 Greenway (2025)

This is a multi million worth of problem that can be solve.

17.38% (2023) → 18.59% (2024)
Of Company Attrition Rate

What can we do

Proactive Approach

Historically, we only learn the employees' experience through conducting **anonymous survey** or conducting regular **exit interviews** to receive feedbacks on the last day of employment (Source 4: Coursera (2024))

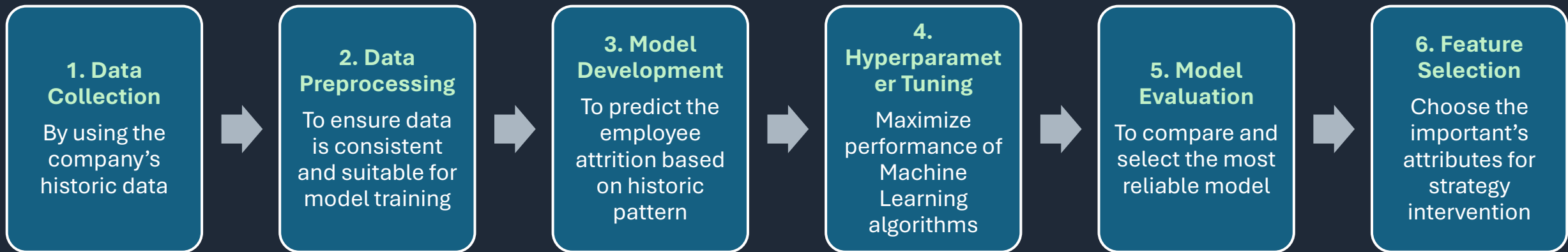
Our solution enable organizations to proactively **retain employees** rather than reactively respond to attrition.

The Goals

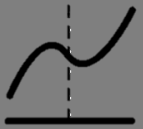
Use Machine Learning techniques to predict who is likely to leave, and act early to retain top talent.



The Methodology

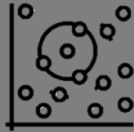


Machine Learning Models for Attrition Prediction



1. Logistic Regression

It is simple, easy to interpret, and fast to train, making it efficient with minimal computational resources.



2. K-Nearest Neighbours

It makes no assumptions about the data distribution and offers flexibility through the choice of the number of neighbors.



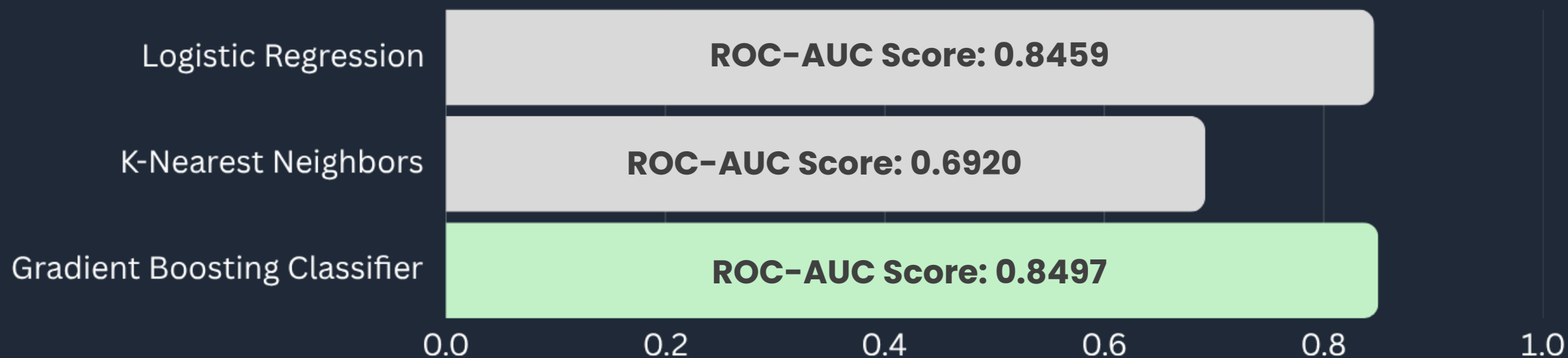
3. Gradient Boosting Classifier

It can handle both numerical and categorical features (through one-hot encoding) and it provides feature importance analysis.



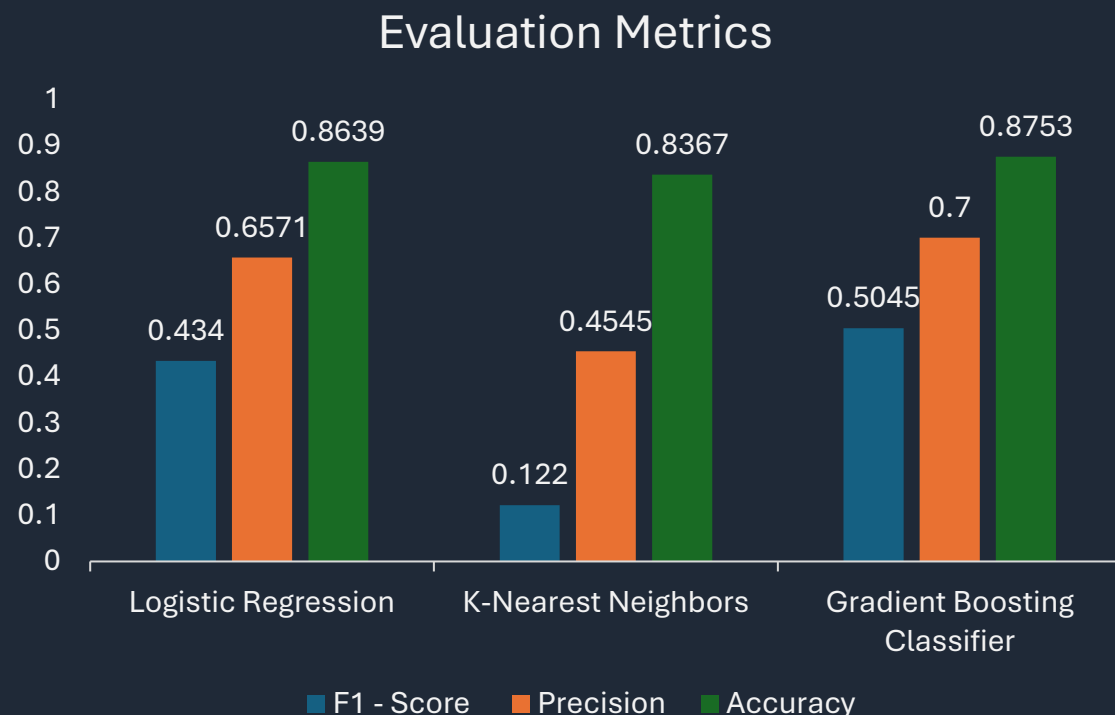
Model Performance & Key Findings

Model Performance (ROC – AUC Score)



Our chosen model, **Gradient Boosting Classifier**, shows the highest ROC-AUC Score (0.8497). This means that the model is highly effective to distinguish between employees who will stay and leave

Other Model Performance (Accuracy, Precision, F1 – Score)



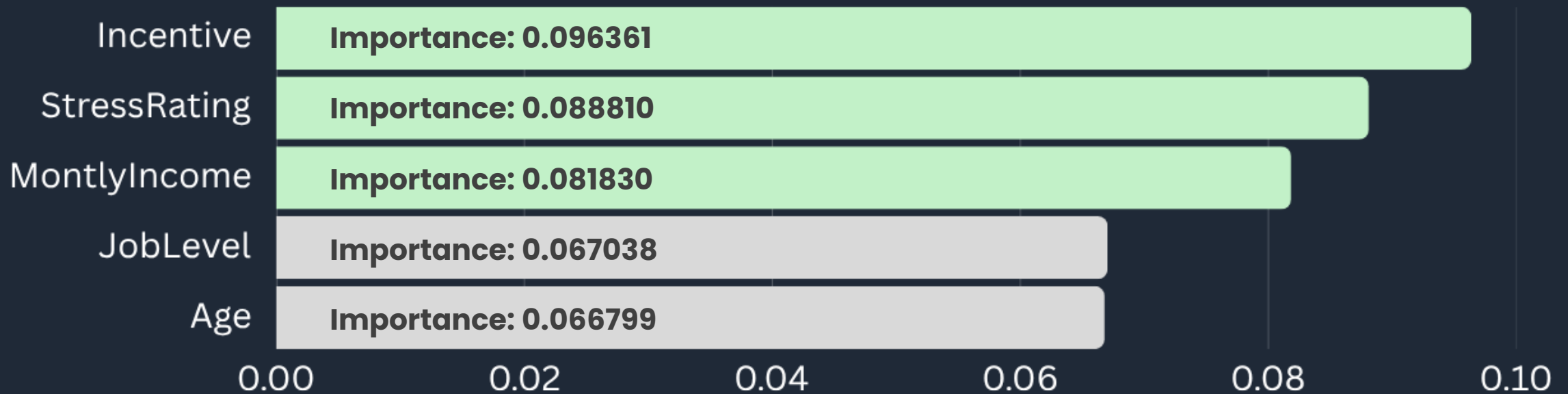
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Logistic Regression
Best Parameters: {'C': 0.1, 'penalty': 'l2'}
F1-Score: 0.4340
Precision: 0.6571
Accuracy: 0.8639
ROC-AUC Score: 0.8459

K-Nearest Neighbors
Best Parameters: {'n_neighbors': 11, 'weights': 'uniform'}
F1-Score: 0.1220
Precision: 0.4545
Accuracy: 0.8367
ROC-AUC Score: 0.6920

Gradient Boosting Classifier
Best Parameters: {'learning_rate': 0.2, 'n_estimators': 200}
F1-Score: 0.5045
Precision: 0.7000
Accuracy: 0.8753
ROC-AUC Score: 0.8497
```

Additional evaluation metrics further validate that the **Gradient Boosting Classifier** is the strongest performing model with the highest accuracy, precision, and F1-score

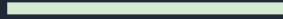
The 'Which': Top Factors Influencing Attrition



What are the next steps?

Based on: Gradient Boosting Classifier

Top 5 important features:		
	Feature	Importance
24	Incentive	0.096361
27	StressRating	0.088810
26	MonthlyIncome	0.081830
8	JobLevel	0.067038
0	Age	0.066799



Recommendations

Proposed Solutions



Financial Categorization

Incentive, Monthly Income, Job Level

- Use the model to flag high performer with underpaid employees that flagged as high risks of attrition for an **proactive salary review**.
- Conduct **incentive reviews** annually to keep incentive pay aligned



Well-being Categorization

Stress Rating, Age

- Offer **flexible work arrangement** in working hours or allowing employees for working remotely (Source 6: PayCaptain (2025))
- Analyze age-related attrition patterns further using ML techniques to take further actions

References

Source 1: <https://www.aihr.com/blog/attrition-rate/>

Source 2: <https://www.bamboohr.com/blog/employee-turnover-vs-attrition-context-is-the-key>

Source 3: <https://payfit.com/blog/cost-of-employee-turnover/>

Source 4:

https://www.coursera.org/enterprise/articles/understanding-attrition?utm_medium=sem&utm_source=gg&utm_campaign=b2c_emea_x_multi_ftcof_career-academy_cx_dr_bau_gg_pmax_gc_gb_en_m_hyb_25-04_x&campaignid=22490486852&adgroupid=&device=c&keyword=&matchtype=&network=x&devicemodel=&creativeid=&assetgroupid=6577627676&targetid=&extensionid=&placement=&gad_source=1&gad_campaignid=22483915359&gbraid=0AAAAADdKX6bM1ugW4QmivHZ8Z2m5GsSdl&gclid=Cj0KCQiAiKzIBhCOARIsAKpKLAPUSdkbZk2CMs-aoV5l7nmfvidEniDKa41ep33QKI6Tw7i-KPqQk6YaAobuEALw_wcB

Source 5:

https://www.paycaptain.com/post/workplace-stress-how-to-recognise-the-signs-and-symptoms-in-employees?gad_source=1&gad_campaignid=23233100152&gbraid=0AAAAACe0RllV6ZivpdeKMbYPjXvyZZSwe&gclid=Cj0KCQiAq7HIBhDoARIsAOATDxBISsTPpnY6HfPvtUKEPkjP0FZq4ieJMGH5CJwPguaph0763eq3OEEaAiT2EALw_wcB